

# A Machine Learning-Based Wearable System for Real-Time Fatigue Detection in Factory Workers

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**Abstract**—Fatigue is a serious issue for the safety of factory workers, which can affect their ability to function physically, mentally and safely on the job. Existing fatigue monitoring systems on the wearer have complex architectures comprising multiple sensors, or cloud-based processing, making them impractical for use in real-time on the shop floor in industrial applications. This piece of work presents a wearable compact system using ESP32 microcontroller along with feature engineering and machine learning (Logistic Regression, Random Forest, HistGB, and SVM) to perform the classification of Fatigue from a 35-minute session with only button labels. Random Forest showed the best performance on the held-out test episodes (F1 = 0.786, accuracy = 0.905, ROC-AUC = 0.939) but was not the highest scorer on cross validation, and HistGB, which had the highest cross validation score, generalized poorly (F1 = 0.222), which shows that it had overfitted, mostly due to the PPG heart rate features. The results here show the potential of the multimodal wearable fatigue detection system, but also suggest the need for more extensive, multi-subject datasets and accurate ground-truth data to enable the system to be deployed effectively in real factory environments.

**Index Terms**—wearable sensors, fatigue detection, machine learning, esp32, physiological signal monitoring, occupational safety

## I. INTRODUCTION

In the field of workplace safety fatigue is one of the major concerns, particularly for the factory workers who repeatedly undertake heavy loads, difficult tasks and working in difficult conditions. Fatigue can reduce the physical productivity, mental concentration and safety of the person in the workplace. Previous studies have found that certain workplace factors (e.g., workload and workplace conditions) are linked to worker fatigue, necessitating the need for appropriate fatigue monitoring methods [1, 2].

A promising approach is to use the wearable sensing and machine learning to continuously monitor fatigue, by combining physiological and motion related signals. Parameters that can be measured with wearable technologies are, for example, heart rate and inertial movements, while machine learning can be applied to identify patterns related to physical fatigue [3, 4]. It has also been demonstrated that these physiological monitoring systems can be used for a variety of occupational groups [5]. Yet most existing systems are complicated by the

need for expensive hardware, cloud computing and tightly controlled environments, which are impractical in factories, and high cost and delays to implement.

In this project, we want to develop a compact, low cost, wearable system for fatigue detection utilizing motion and physiology sensors and machine learning on-board. The system is designed to continuously track workers, dynamically categorize real-time fatigue level and give local alerts, relay fatigue information to supervisors, and keep the system low cost, low power consumption, and user comfortable.

The primary goals of this work are to perform real-time motion and physiological signal processing with ESP32 using MPU6050, MAX30102 and AD8232 sensors, implement a lightweight SVM/1D-CNN model to detect fatigue, design a buzzer/LED alert system with BLE and Wi-Fi dashboard connectivity and evaluate the system in realistic scenarios to assess the system accuracy, battery life, comfort, response time, and hardware cost.

The rest of this paper is organized as follows. Related work is discussed in Section II. Details of our data and model setup are described in Section III. The results of the tests are presented in section IV. Section V points out limitations, and Section VI concludes the study.

## II. RELATED WORKS

A summary of the related works is given in Table I. Sedighi Maman et al. [6] presented a data-analytic framework of physical fatigue management with wearable sensors. For manufacturing-related cases their approach had an average fatigues prediction accuracy of at least 85%. The most beneficial sensors and features were, however, somewhat task-dependent and thus were not easily generalizable.

Liu et al. [7] suggested a machine-learning method to detect fatigue in repetitive physical activities based on wearable motion, physiological and force measurement. They obtained the best polynomial-kernel SVM with a maximum accuracy of 94.15%. The system needed several sensing elements and was primarily tested on structured tasks, however.

In order to analyze eye fatigue for field workers, Lee et al. [8] proposed a computer-vision based eye fatigue analysis system. It provided the workers' safety, by allowing real

TABLE I: Summary of selected related studies on wearable fatigue detection

| Ref. | Year | Method                                  | Results                       | Limitations                  |
|------|------|-----------------------------------------|-------------------------------|------------------------------|
| [6]  | 2020 | Wearable sensors + data analytics       | $\geq 85\%$ accuracy          | Limited generalization       |
| [7]  | 2023 | Motion, physiological, force data + SVM | 94.15% accuracy               | Repetitive tasks only        |
| [8]  | 2023 | Computer vision (eye fatigue)           | Real-time detection           | Lighting/occlusion sensitive |
| [9]  | 2024 | Multi-sensor wearable network + ML      | Predicted fatigue levels      | High hardware complexity     |
| [10] | 2025 | IMU sensors + ML                        | Fatigue trends (construction) | Limited physiological data   |
| [11] | 2026 | Systematic review (wearables + ML)      | Strong safety potential       | High method variation        |
| [12] | 2026 | AI-based HRV monitoring                 | Early fatigue detection       | Needs real-world validation  |

time fatigue analysis. Nonetheless, lighting, viewing angle, occlusion, and computational requirements can all impact camera-based methods.

In order to predict multiple levels of physical fatigue among the manufacturing workers, Mohapatra et al. [9] proposed a wearable network. Their findings showed that it's possible to integrate several wearables signals to predict fatigue. The multi-sensor architecture, however, makes the hardware more complex and system more costly.

Keränen et al. [10] suggested using IMU wearable sensor data to endorse machine-learning based fatigue trend analysis for construction workers. Their findings indicated the possibility of using IMU signals for trend recognition in a realistic working environment during states of fatigue. The study was mainly focused on the motion sensing, however, no attempt was made to integrate the information from the IMU with information from the physiological sensing in a compact system.

Peters et al. [11] and Rusana et al. [12] reviewed the wearable sensors and machine-learning algorithms used for occupational safety, and AI-driven wearable HRV monitoring for early fatigue detection in nurses, respectively. Both identified significant opportunities to use these technologies that would place sensors on workers and use AI to forecast their tiredness early on, for greater safety. Comparisons between studies, however, are restricted due to differences in the sensors and protocols used and require additional field testing in the workplace.

### III. METHODOLOGY

#### A. System Implementation and Machine Learning Pipeline

Data were collected with an ESP32-based wearable combining a MAX30102 PPG sensor (heart rate, skin temperature), an MPU6050 IMU (motion), and an AD8232 single-lead ECG front end (heart rate, HRV), all wired directly to the ESP32's native 3.3 V ADC/GPIOs (Figure 1). A push button let the subject self-mark fatigue events, giving the ground-truth label used for training. Figure 2 represents the practical implementation of the prototype.

The ESP32 samples ECG at  $\sim 250$  Hz and detects R-peaks with an adaptive baseline (Listing 1), from which ECG heart rate and HRV are derived; PPG heart rate comes from the MAX30102 library. A 1 Hz summary (all sensor values + button label) was logged for training; the on-device buzzer

alert uses a fixed threshold rule, independent of the classifier below.

#### Listing 1: ECG R-peak detection.

```

void detectEcgPeak(int sample) {
    ecgBaseline = ecgBaseline*0.995 + sample*0.005;
    ecgSignalValue = abs((int)(sample - ecgBaseline));
    if (ecgSignalValue > ECG_PEAK_THRESHOLD
        && !ecgAboveThreshold){
        ecgAboveThreshold = true;
    }
    if (now - ecgLastPeakTime > ECG_REFRACTORY_MS){
        unsigned long rr = now - ecgLastPeakTime;
        if (rr > 300 && rr < 2000){
            ecgBpm = 60000 / rr; recordRR(rr);
        }
        ecgLastPeakTime = now;
    }
}

```

The heart rate, HRV and motion metrics are continuously computed from the MAX30102, MPU6050 and AD8232, as illustrated in Figure 3. Local and optional Telegram alerts are activated by fatigue thresholds and readings are uploaded to the cloud. This process is repeated about every 4,ms.

One 35.2-minute session yielded 2,004 one-second rows and 91,530 raw ECG samples ( $\sim 43.3$  Hz), with 47 button-marked episodes (23 fatigued, 24 rest). Episodes, not rows, were used as the grouping unit for splitting, to prevent leakage. Raw ECG was aggregated per row into windowed statistics (mean, std, peak count, etc.), combined with PPG/ECG heart rate, HRV, motion, temperature, 5-sample rolling means/stds, and warm-up flags 28 features total. As shown in Figure 4, episodes were split 75/25 (35 train / 12 test episodes; 1,624 / 380 rows). Logistic regression, random forest, HistGB, and SVM were tuned via GridSearchCV with grouped 5-fold CV (episode-level groups), scored on F1 for the fatigued class, then evaluated once on the held-out test episodes.

### IV. RESULTS AND DISCUSSION

TABLE II: Grouped 5-fold CV F1 (mean  $\pm$  std).

| Model        | CV F1 mean | CV F1 std |
|--------------|------------|-----------|
| LogReg       | 0.367      | 0.308     |
| RandomForest | 0.454      | 0.374     |
| HistGB       | 0.458      | 0.376     |
| SVM          | 0.373      | 0.309     |

As shown in Table II, HistGB achieved the highest mean F1-score (0.458) among the evaluated models, followed closely

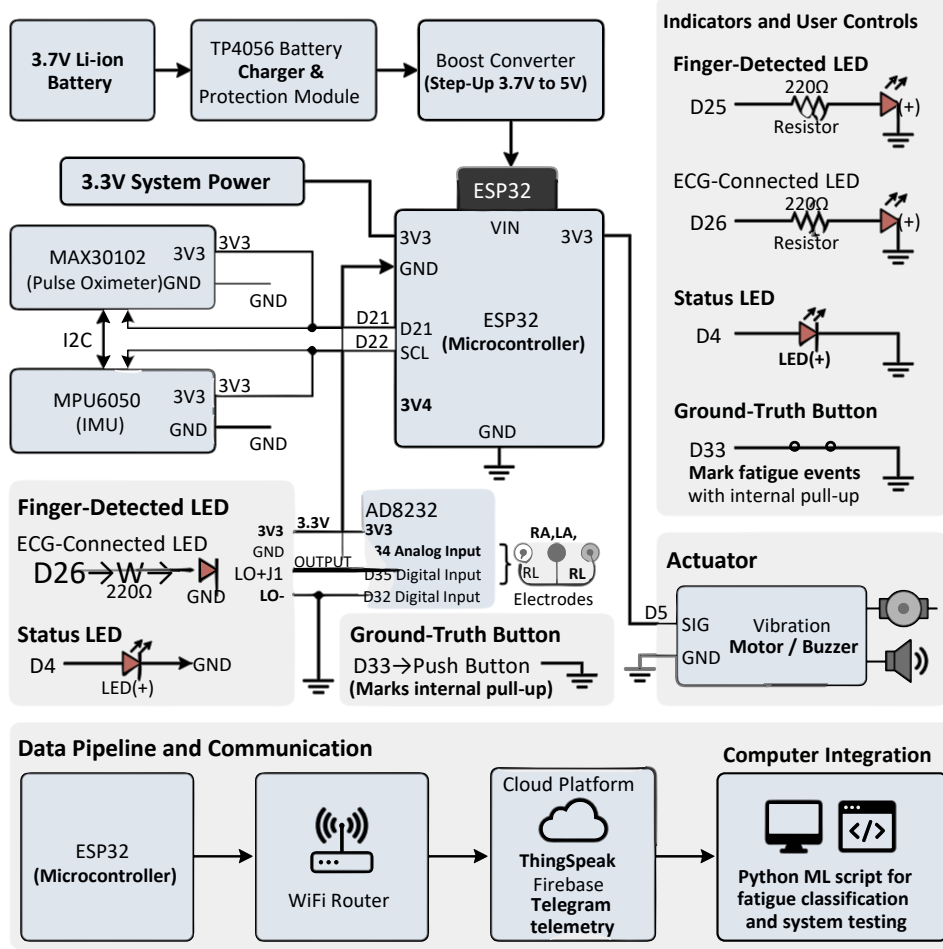


Fig. 1: Wiring diagram of the ESP32 wearable prototype.

by Random Forest (0.454). In comparison, SVM and Logistic Regression achieved mean F1-scores of 0.373 and 0.367, respectively. The relatively large standard deviations observed across all models suggest substantial variation in performance among the validation folds.

TABLE III: Held-out test performance (fatigued class).

| Model        | Precision | Recall | F1           | Accuracy | ROC-AUC |
|--------------|-----------|--------|--------------|----------|---------|
| RandomForest | 0.767     | 0.805  | <b>0.786</b> | 0.905    | 0.939   |
| LogReg       | 0.661     | 0.902  | 0.763        | 0.879    | 0.957   |
| SVM          | 0.680     | 0.622  | 0.650        | 0.855    | 0.879   |
| HistGB       | 0.647     | 0.134  | 0.222        | 0.797    | 0.931   |

As shown in Table III, Random Forest achieved the best overall performance (F1 = 0.786, accuracy = 0.905, ROC-AUC = 0.939), offering the strongest precision-recall balance. Logistic Regression achieved the highest recall (0.902) and ROC-AUC (0.957), while HistGB, despite the top cross-validation score, generalized poorly (F1 = 0.222, recall =

0.134) evident in its confusion matrix and ROC/PR curves (Figures 5a and 6) and confirmed by a persistent 0.49 train/CV F1 gap (Figure 5b), suggesting its CV-tuned parameters overfit the training folds. Permutation importance showed predictions were dominated by rolling-mean and instantaneous PPG BPM (0.483 and 0.377), with ECG and motion features contributing almost nothing. Figure 7 shows raw sensor streams (PPG BPM, ECG BPM, ECG HRV, motion) across the session, with fatigued periods shaded to allow visual correlation with self-reported fatigue.

## V. LIMITATIONS AND FUTURE WORKS

This was limited to one subject and 35 minutes of study, which decreased the generalizability of the study. The model performance was unstable and test-set bias and reliance on self-reported fatigue were possible. Heart-rate features dominated predictions and ECG signals were influenced by motion and electrode contact. To enhance the study and validation of fatigue metrics, larger multi-subject datasets should be

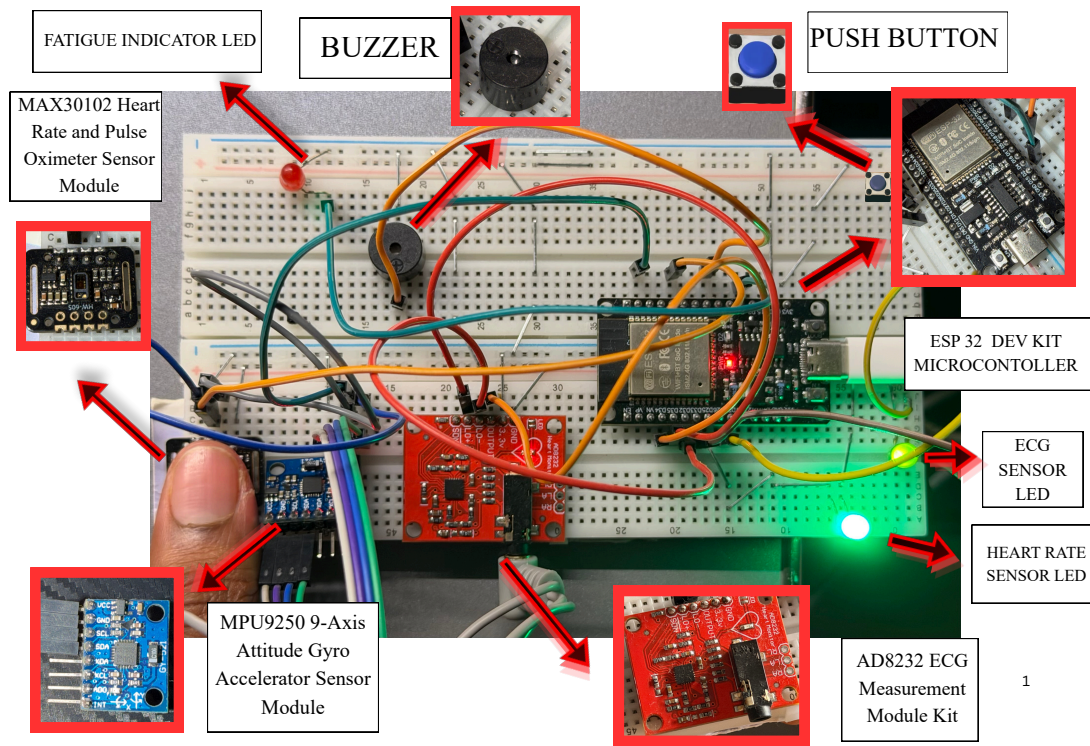


Fig. 2: Hardware Implementation of the prototype

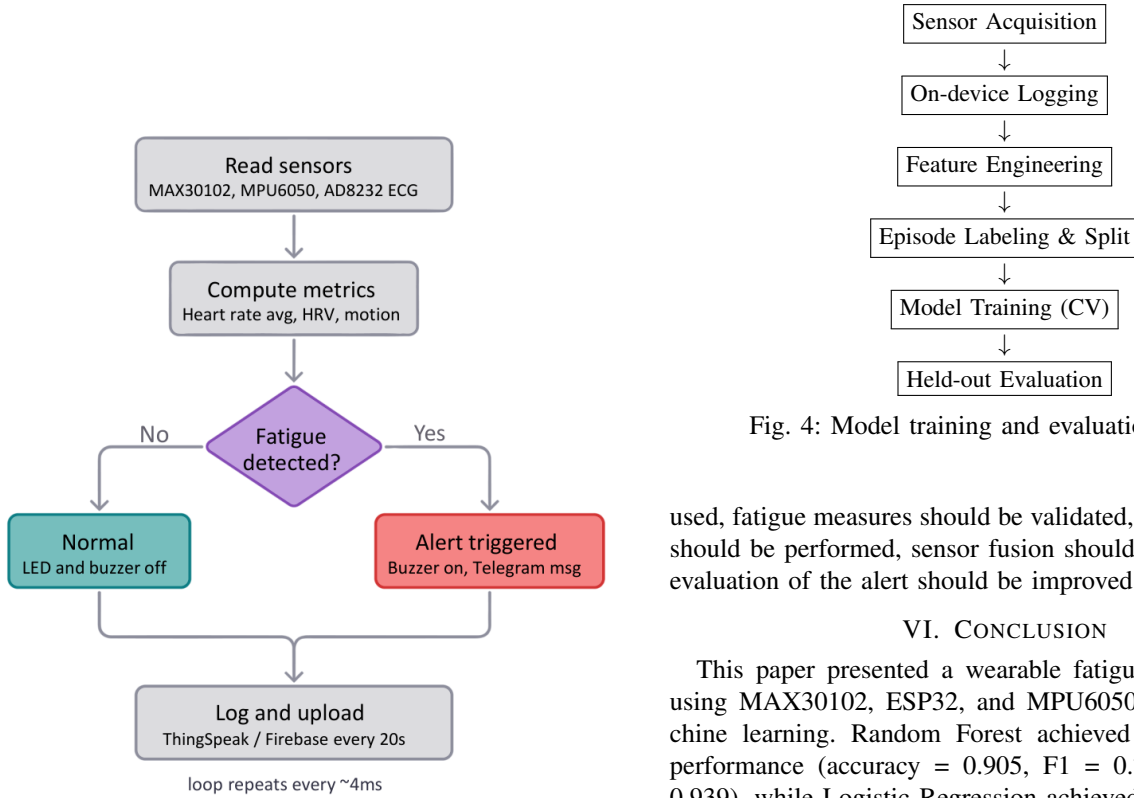


Fig. 3: Workflow diagram

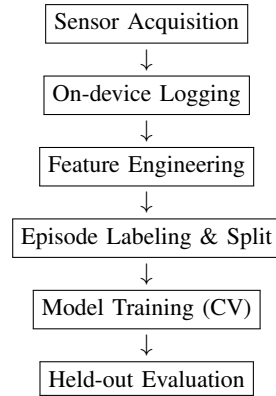


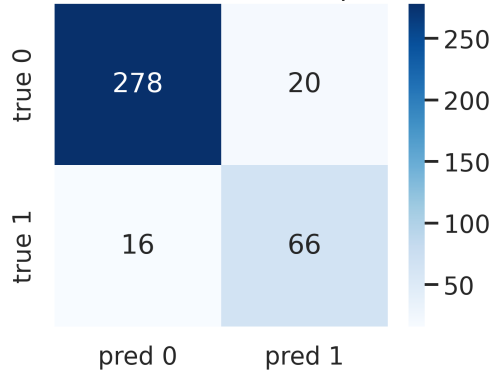
Fig. 4: Model training and evaluation pipeline.

used, fatigue measures should be validated, external validation should be performed, sensor fusion should be optimized, and evaluation of the alert should be improved in future research.

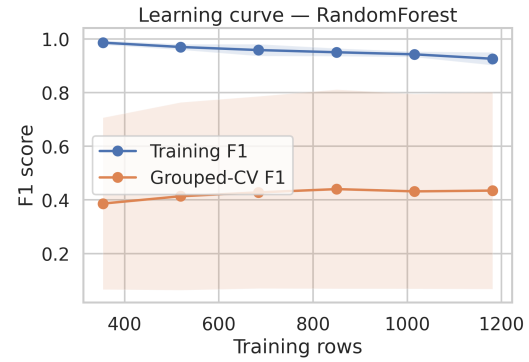
## VI. CONCLUSION

This paper presented a wearable fatigue-detection system using MAX30102, ESP32, and MPU6050 sensors with machine learning. Random Forest achieved the best held-out performance (accuracy = 0.905, F1 = 0.786, ROC-AUC = 0.939), while Logistic Regression achieved the highest ROC-AUC (0.957). However, the single-subject dataset, signs of overfitting, and dominance of heart-rate features indicate that the model may be detecting a rest/heart-rate proxy rather than true multimodal fatigue. Larger multi-subject studies, external

RandomForest (held-out episodes)



(a) Confusion matrix (Random Forest, test set).



(b) Learning curve (Random Forest).

Fig. 5: Random Forest evaluation: confusion matrix and learning curve.

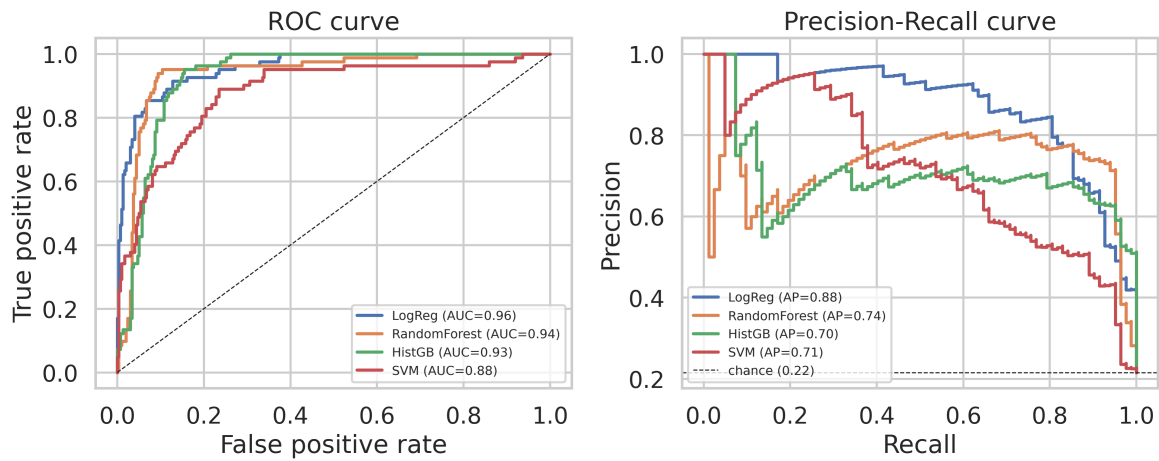


Fig. 6: ROC and precision–recall curves for all models.

validation, and improved sensor fusion are required before deployment in real factory environments.

#### ACKNOWLEDGMENT

The authors are grateful to all the faculty members of the Department of Computer Science, American International University-Bangladesh for their guidance, support and constructive feedbacks during this work.

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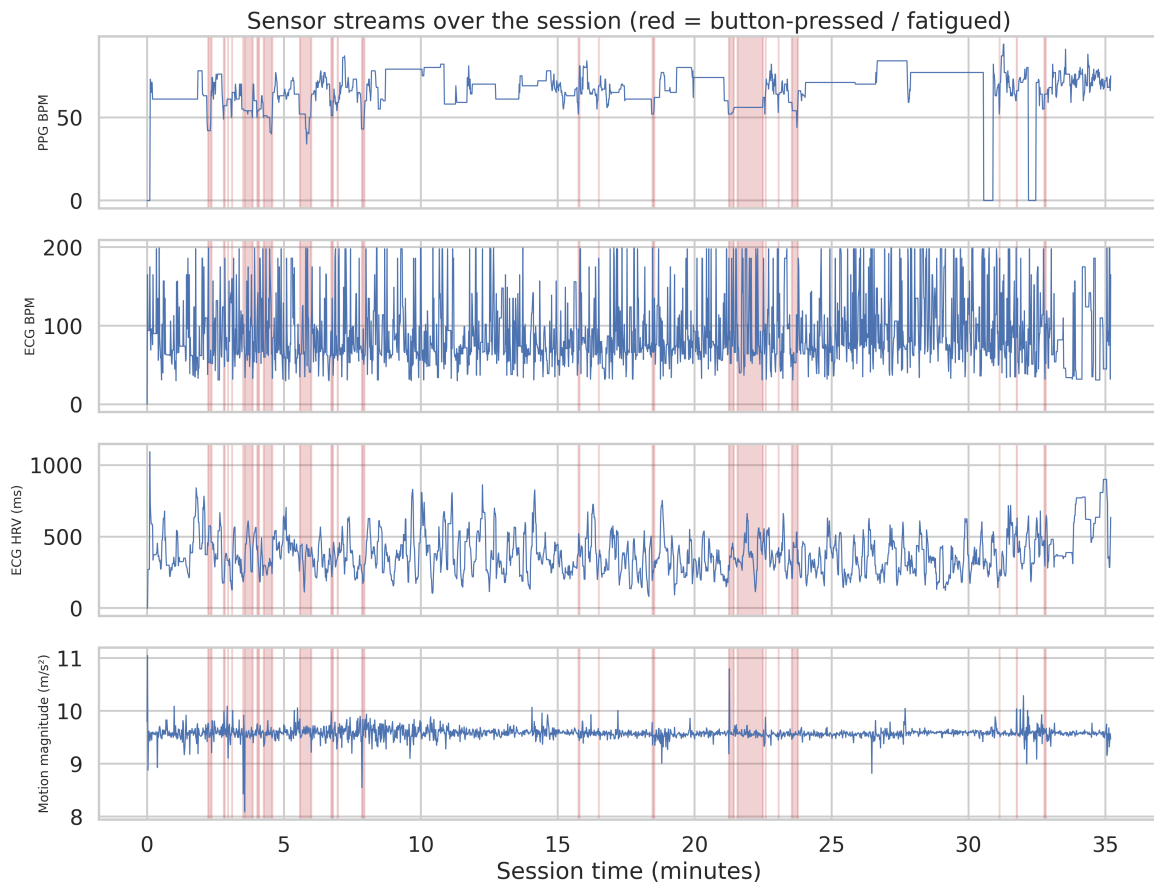


Fig. 7: Sensor streams over the session; shaded regions mark button-pressed (fatigued) periods.

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